School of Engineering and Applied Science (SEAS), Ahmedabad University
CSE 400: Fundamentals of Probability in Computing
Lecture 18 Scribe: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables
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1 Marginal PMF

For discrete random variables X and Y, the joint PMF is defined as $p_{XY}(x, y) = P(X = x, Y = y)$[cite: 73, 74]. The marginal PMFs are obtained by summing the joint PMF over all possible values of the other variable[cite: 75, 78].

1.1 Definitions

- Marginal PMF of X: $p_X(x) = \sum_y p_{XY}(x, y)$ [cite: 76]
- Marginal PMF of Y: $p_Y(y) = \sum_x p_{XY}(x, y)$ [cite: 76]

When working with a Joint PMF Table, the row sums give $p_X(x)$, and the column sums give $p_Y(y)$[cite: 85].

2 Correlation

Correlation is a basic way to measure joint behavior through the product moment[cite: 134].

- In the continuous case, it is defined as:

$$R_{XY} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\ f_{XY}(x, y)dx\ dy$$

  [cite: 137]

- It represents the average value of the product XY ($R_{XY} = E[XY]$)[cite: 136, 138].
- It depends on the scale of X and Y and mixes mean values, spread, and dependence[cite: 144, 146, 147, 148, 149].

3 Covariance

To remove the effect of the means and measure centered dependence, we center the variables to find the covariance[cite: 154].

3.1 Definition and Interpretation

Covariance is defined as:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

[cite: 162] where $\mu_X = E[X]$ and $\mu_Y = E[Y]$[cite: 163].

Covariance tells us the direction of joint linear variation between X and Y[cite: 179].

- **Positive Covariance ($Cov(X,Y) > 0$):** Deviations tend to have the same sign[cite: 172, 175].
- **Negative Covariance ($Cov(X,Y) < 0$):** One tends to increase when the other decreases[cite: 173, 176].
- **Zero Covariance ($Cov(X,Y) = 0$):** No linear co-variation trend[cite: 174, 177].

3.2 Algebraic Form and Properties

The algebraic formula for covariance is:

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

[cite: 229]

Key properties include[cite: 240, 241, 243, 254, 267]:

1. **Symmetry:** $Cov(X,Y) = Cov(Y,X)$ [cite: 242]
2. **Variance as a Special Case:** $Cov(X,X) = Var(X)$ [cite: 244]
3. **Linearity:** $Cov(aX+b, cY+d) = ac\, Cov(X,Y)$ [cite: 255]
4. **Independence:** $X \perp Y \Rightarrow Cov(X,Y) = 0$[cite: 268]. Zero covariance means uncorrelated, but it does not strictly imply independence[cite: 269].

4 Pearson's Correlation Coefficient

The magnitude of covariance depends on the units of the variables[cite: 180]. The Pearson correlation coefficient (ρ_{XY}) is the normalized version of covariance that measures the strength and direction of the linear relationship[cite: 386, 387].

4.1 Formula and Constraints

$$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

[cite: 385]

- It is always between -1 and 1 ($-1 \leq \rho_{XY} \leq 1$)[cite: 355, 398].
- It is unit-free[cite: 388].
- $\rho_{XY} > 0$ indicates a positive linear trend, $\rho_{XY} < 0$ indicates a negative linear trend, and $\rho_{XY} = 0$ indicates no linear correlation[cite: 399, 400]. Even if $\rho_{XY} = 0$, a non-linear relationship may still exist[cite: 377].

5 Jointly Gaussian Random Variables

A pair (X, Y) is jointly Gaussian if its joint density has the bivariate Gaussian form and both marginal distributions are Gaussian ($X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$)[cite: 569, 570].

5.1 Algebraic Form

For two variables X and Y, the Joint Gaussian PDF is[cite: 591]:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where:

- μ_X, μ_Y = means [cite: 593]
- σ_X, σ_Y = standard deviations [cite: 594, 595]
- ρ = Pearson correlation coefficient [cite: 596, 597]